



## Lab Report - 03

Experiment on 03

Course Name: Structured Programming Lab

Course Code: CSE-122

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Submission Date: 2025-07-03

## Lab Report-03

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Signature:

Date :

**Problem-01:** Write a C program to read the value of an integer m and display the value of n is 1 when m is larger than 0, 0 when m is 0 and -1 when m is less than 0.

Test Data :-5

Expected Output:

The value of n =-1

**Source code:**

```
e X Untitled1.c X
1  #include<stdio.h>
2  int main()
3  {
4      int m;
5      scanf("%d",&m);
6      if (m>0)
7          printf("The value of n =1");
8      else if (m=0)
9          printf("The value of n=0");
10     else
11         printf("The value of n=-1\n");
12
13     return 0;
14
15 }
```

**Output:**

```
-5
The value of  n=-1

Process returned 0 (0x0)   execution time : 3.597 s
Press any key to continue.
```

**Problem-02:** Write a C program to accept the height of a person in centimeters and categorize the person according to their height where height less than 150cm is considered a dwarf, greater than 160cm is considered as tall and a height of 150cm is considered as an average height person.

Test Data :

135

Expected Output:

The person is Dwarf.

**Source code:**

```
re X  Untitled1.c X
1  #include<stdio.h>
2  int main()
3  {
4      int h;
5      scanf("%d",&h);
6      if(h<150)
7          printf("The person is Dwarf.\n");
8      else if(h>160)
9          printf("The person is Tall.\n");
10     else
11         printf("The person is Average.\n");
12
13     return 0;
14
15 }
```

**Output:**

```
135
The person is Dwarf.

Process returned 0 (0x0)   execution time : 3.107 s
Press any key to continue.
|
```

**Problem-03:** Write a C program to accept a coordinate point in an XY coordinate system and determine in which quadrant the coordinate point lies.

Test Data :

7 9

Expected Output:

The coordinate point (7,9) lies in the First quadrant.

**Source code:**

```
iere x  Untitled1.c x
1  #include<stdio.h>
2  int main()
3  {
4      int x,y;
5      scanf("%d%d",&x,&y);
6      if(x>0,y>0)
7          printf("The coordinate point (%d,%d) lies in the First quadrant.\n",x,y);
8      else if (x<0,y>0)
9          printf("The coordinate point (%d,%d) lies in the Second quadrant.\n",x,y);
10     else if (x<0,y<0)
11         printf("The coordinate point (%d,%d) lies in the Third quadrant.\n",x,y);
12     else if (x>0,y<0)
13         printf("The coordinate point (%d,%d) lies in the Forth quadrant.\n",x,y);
14
15     return 0;
16
17 }
```

**Output:**

```
7 9
The coordinate point (7,9) lies in the First quadrant.

Process returned 0 (0x0)    execution time : 3.786 s
Press any key to continue.
|
```

**Problem-04:** Write a C program to read the temperature in centigrade and display a suitable message according to the temperature state below:

Temp < 0 then Freezing weather

Temp 0-10 then Very Cold weather

Temp 10-20 then Cold weather

Temp 20-30 then Normal in Temp

Temp 30-40 then It's Hot

Temp >=40 then It's Very Hot

Test Data :

42

Expected Output:

It's Very Hot.

**Source code:**

```
e X  Untitled 32.c X
1  #include<stdio.h>
2  int main()
3  {
4      int x;
5      scanf("%d",&x);
6      if (x<0)
7          printf("\nFreezing weather.\n");
8      else if(x>=0 && x<=10)
9          printf("\nVery Cold weather.\n");
10     else if (x>10 && x<=20)
11         printf("\nCold weather.\n");
12     else if (x>=20 && x<=30)
13         printf("\nNormal in Temp.\n");
14     else if (x>=30 && x<=40)
15         printf("\nIt's Hot.\n");
16     else
17         printf("\nIt's Very Hot.\n");
18
19     return 0;
20 }
```

## Output:

42

It's Very Hot.

Process returned 0 (0x0)    execution time : 2.896 s

Press any key to continue.

|

**Problem-05:** Write a C program to check whether a triangle is Equilateral, Isosceles or Scalene.

Test Data :

50 50 60

Expected Output:

This is an isosceles triangle.

**Source code:**

```
re X Untitled 30.c X Untitled31.c X
1  #include <stdio.h>
2  int main()
3  {
4      int x,y,z;
5      scanf("%d%d%d",&x,&y,&z);
6      if (x==y && y==z && z==x )
7          printf("This is an equilateral triangle.\n ");
8      else if (x==y || y==x || z==x)
9          printf("This is an isosceles triangle.\n");
10     else
11         printf("This is a scalene triangle.\n");
12
13     return 0;
14
15 }
```

**Output:**

```
50 50 60
```

```
This is an isosceles triangle.
```

```
Process returned 0 (0x0)   execution time : 4.900 s
Press any key to continue.
```



**Problem-06:** Write a C program to check whether an alphabet is a vowel or a consonant.

Test Data :

P

Expected Output:

The alphabet is a consonant.

**Source code:**

```
1  #include<stdio.h>
2  int main()
3  {
4      int ch;
5      scanf("%c",&ch);
6      if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
7          printf("\nThe alphabet is an vowel.\n");
8      else
9          printf("\nThe alphabet is a consonant.\n");
10
11     return 0;
12
13 }
```

**Output:**

```
p

The alphabet is a consonant.

Process returned 0 (0x0)   execution time : 2.381 s
Press any key to continue.
```

**Problem-07:** Write a C program to calculate profit and loss on a transaction.

Test Data :

Enter buying price: 500

Enter selling price: 300

Expected Output:

Loss amount: 200

**Source code:**

```
ere X Untitled2.c X
1  #include<stdio.h>
2  int main()
3  {
4
5      int buy,sell,profit,loss;
6      printf("Enter buying price:");
7      scanf("%d",&buy);
8      printf("Enter selling price:");
9      scanf("%d",&sell);
10     if (buy>sell)
11         printf("Loss amount:%d\n",buy-sell);
12     else
13         printf("Profit amount:%d\n",sell -buy);
14
15     return 0;
16
17 }
```

**Output:**

```
Enter buying price:500
Enter selling price:300

Loss amount:200

Process returned 0 (0x0)   execution time : 4.187 s
Press any key to continue.
|
```

**Problem-08:** Write a program in C to read any Month Number in an integer and display the number of days for this month.

Test Data :

7

Expected Output :

The month of July has 31 days.

**Source code:**

```
re x Untitled 32.c x
1  #include<stdio.h>
2  int main()
3  {
4      int month;
5      scanf("%d",&month);
6      if (month==1) printf("The month of January has 31 days.\n ");
7      else if (month==2) printf("The month of February has 28 or 29 days.\n ");
8      else if (month==3) printf("The month of March has 31 days.\n ");
9      else if (month==4) printf("The month of April has 30 days.\n ");
10     else if (month==5) printf("The month of May has 31 days.\n ");
11     else if (month==6) printf("The month of June has 30 days.\n ");
12     else if (month==7) printf("The month of July has 31 days.\n ");
13     else if (month==8) printf("The month of August has 31 days.\n ");
14     else if (month==9) printf("The month of September has 30 days.\n ");
15     else if (month==10) printf("The month of October has 31 days.\n ");
16     else if (month==11) printf("The month of November has 30 days.\n ");
17     else
18         printf("The month of December has 31 days.\n ");
19     return 0;
20 }
```

**Output:**

```
7
The month of July has 31 days.

Process returned 0 (0x0)   execution time : 2.335 s
Press any key to continue.
|
```

**Problem-09:** Write a program in C to convert temperature from Fahrenheit to Celsius and Celsius to Fahrenheit.

Test Data :

Press c to convert the temperature from Fahrenheit to Celsius.

Press f to convert temperature from Celsius to Fahrenheit.

Enter your choice (c, f): f

Enter temperature in Celsius: 37

Expected Output :

Temperature in Fahrenheit: 98.60

**Source code:**

```
re x Untitled 32.c x
1  #include<stdio.h>
2  int main()
3  {   char choice;
4      float c,f;
5      printf("Press c to convert the temperature from Fahrenheit to Celsius.\n");
6      printf("press f to convert temperature from Celcius to Fahrenheit.\n");
7      printf("Enter your choice (c,f):");
8      scanf("%c",&choice);
9      if (choice=='c')
10     {printf("Enter temperature in Farhenheit:"); scanf("%f",&f);
11       c=((f-32)*5)/9;
12       printf("\nTemperature in Celsius:%.2f\n",f);}
13     else if(choice=='f')
14     {printf("Enter temperature in Celsius:"); scanf("%f",&c);
15       f=(1.8*c)+32;
16       printf("\nTemperature in Fahrenheit:%.2f\n",f);}
17     else
18       printf(" Invalid choice:\n");
19
20     return 0;
21 }
```

**Output:**

```
Press c to convert the temperature from Fahrenheit to Celsius.
press f to convert temperature from Celcius to Fahrenheit.
Enter your choice (c,f):f
Enter temperature in Celsius:37

Temperature in Fahrenheit:98.60

Process returned 0 (0x0)   execution time : 15.485 s
Press any key to continue.
|
```